

LENS AND TELESCOPE

Objectives

1. To relate image and object distances to focal lengths for various lenses. To be able to determine focal length, image size, and f-numbers of the lenses.
2. To construct a simple optical refracting telescope and measure the angular magnification of the telescope.

Equipment and supplies:

Optical rail with 2 carriages, flashlight, metal plate with an arrow-shaping opening, viewing screen, ruled target, converging lens with 30 cm focal length, converging lens with 5 cm focal length, lens mounts, meter ruler stick.

Introduction:

An astronomical telescope amplifies light with a system of lenses and/or mirrors. A converging lens is defined as a lens with two convex surfaces (thicker in the middle). The focal length of a converging lens is defined as the distance from the lens at which a parallel ray of light impinging on the lens parallel to the optic axis will converge. Converging lenses have two focal points. See the diagram below.

Converging Lens:

Note: Converging lenses are thicker in the middle than at the edges.

Lenses have two focal points, F , symmetrically placed, one on each side. Their distance to the lens is the focal distance, f . Light from a very distant object will converge to a real image in the focal plane as shown in Figure 1.

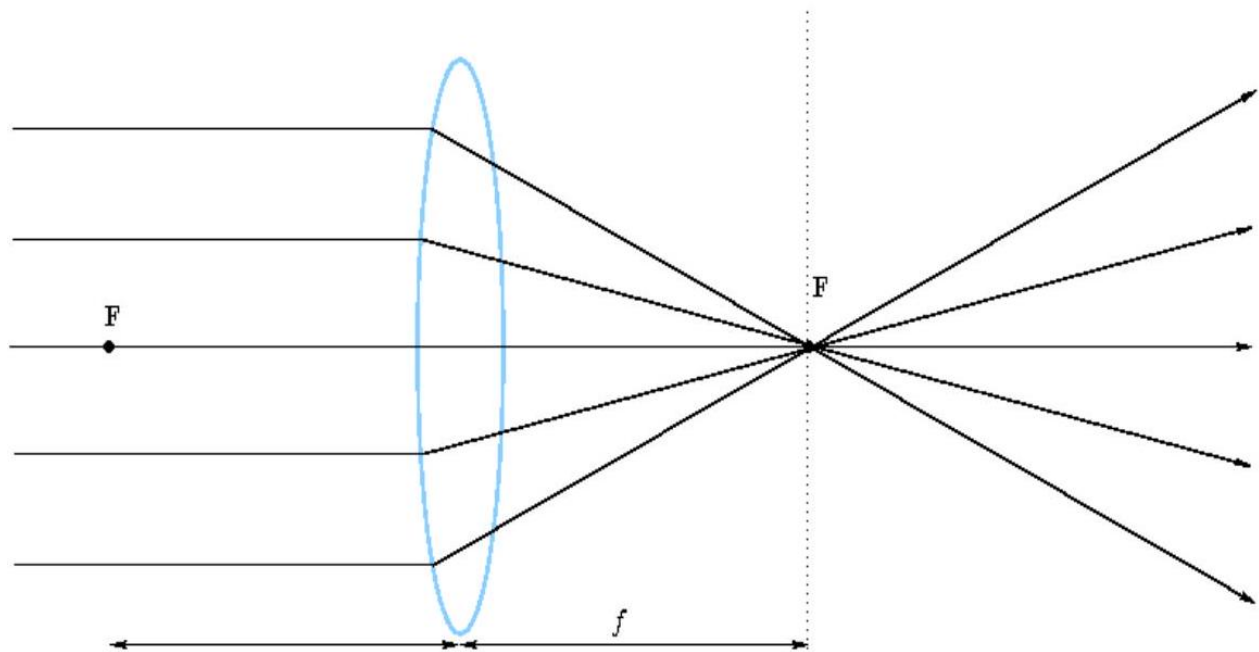


Fig. 1-Light from infinity converges to the focal point.

If the object is not at infinity, the properties of the image depend on the distance of the object to the lens.

1. An object is any article viewed through a optical system. An image of an object is produced in space by an optical system such as a lens. The object distance, p , is the distance from the object to the optical element (lens). Similarly the image distance, q , is the distance from the image to the optical element

(lens). A real object placed at a distance greater than the focal length, will result in an inverted image on the other side of the lens, as shown in Fig. 2. To find the position of the image, we use the ray tracing method. The ray parallel to the axis changes direction and passes through the focal point. The ray passing through the center of the lens does not change direction. The intersection of these two rays is the image of the object. Because the rays actually strike on the image, the image is called a real image.

The position of the image can be predicted using the following relation:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

where p is the object distance, q is the image distance, and f is the focal length.

2. A real object placed between the focal point and the lens, will result in a virtual erect image on the same side of the lens. Following the rules of ray tracing, you can see that the outgoing rays from the object never actually intersect at a point, as shown in Fig. 3. The virtual rays (the dashed lines) represent perceived paths as seen by an observer looking into the optical device. The light rays do not travel on these dotted paths. The virtual image is located where the virtual rays intersect.

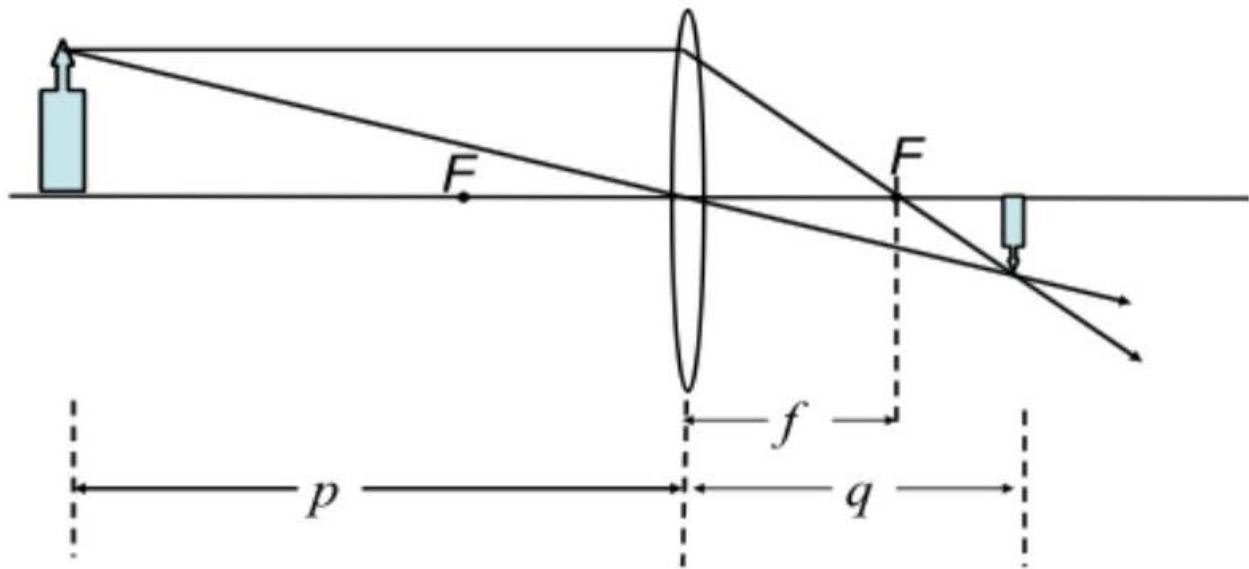


Fig. 2-An object placed at a distance larger than the focal length results in an inverted image.

From Fig. 3, the size of the virtual image is larger than that of the object. Thus when the object is placed between the focal point and the lens, the lens is a magnifier.

Optical Telescope

An optical telescope is an instrument designed for the observation of remote objects by providing angular magnification. Typically, a telescope consists of an objective with a longer focal length and an eyepiece with a shorter focal length. The separation of the lenses is approximately the sum of the focal lengths of these two lenses. A real inverted image (Image 1) is first formed by the objective lens. The eyepiece then forms the magnified image (Image 2) of Image 1. The angular magnification of a telescope is defined as the ratio of angle subtended by the magnified image as viewed through the eyepiece and the angle subtended by the object without the telescope. For a distant object, the angular magnification is given by

$$\text{Magnification} = F_o/F_e$$

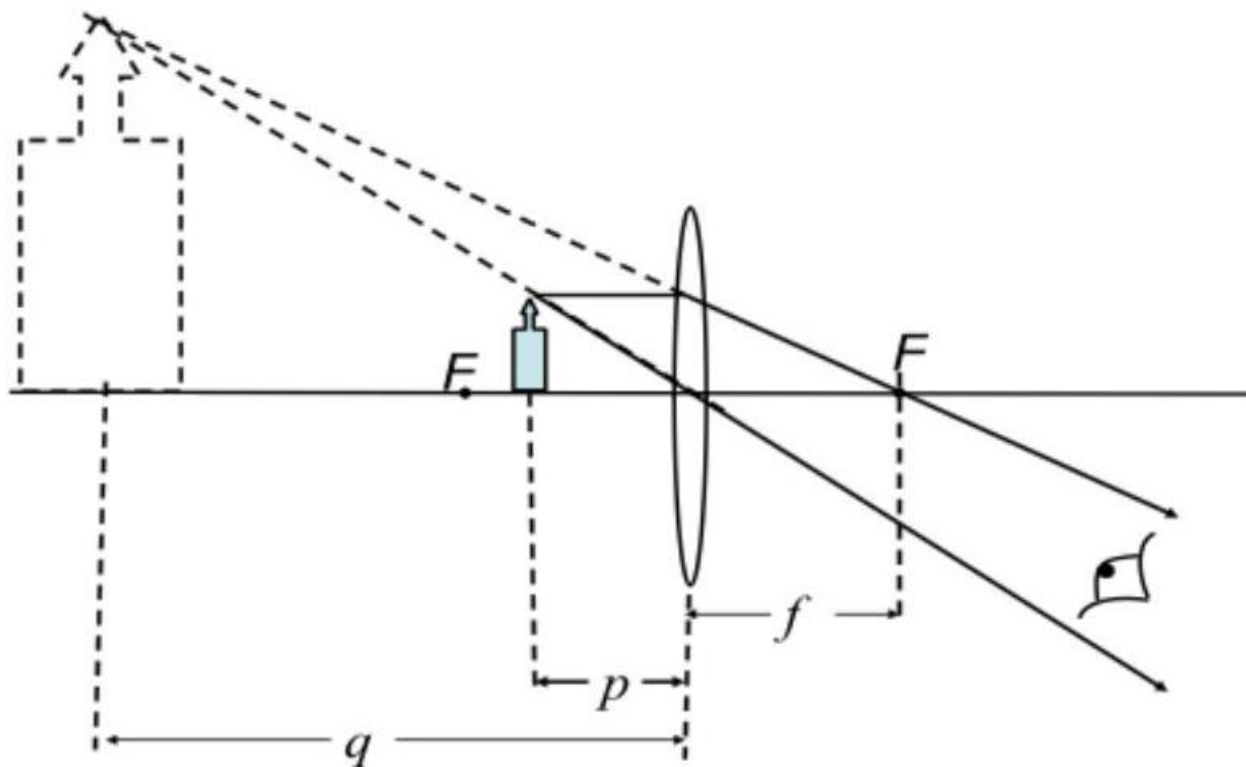


Fig.3. A virtual image is formed when the object is placed between the focal point and the lens.

where F_o and F_e are the focal lengths of the objective and eyepiece, respectively.

Procedure

Two lenses are provided in this experiment. Remove them from the optical rail and place them on the lab bench.

A1: Measure the focal length of the larger lens

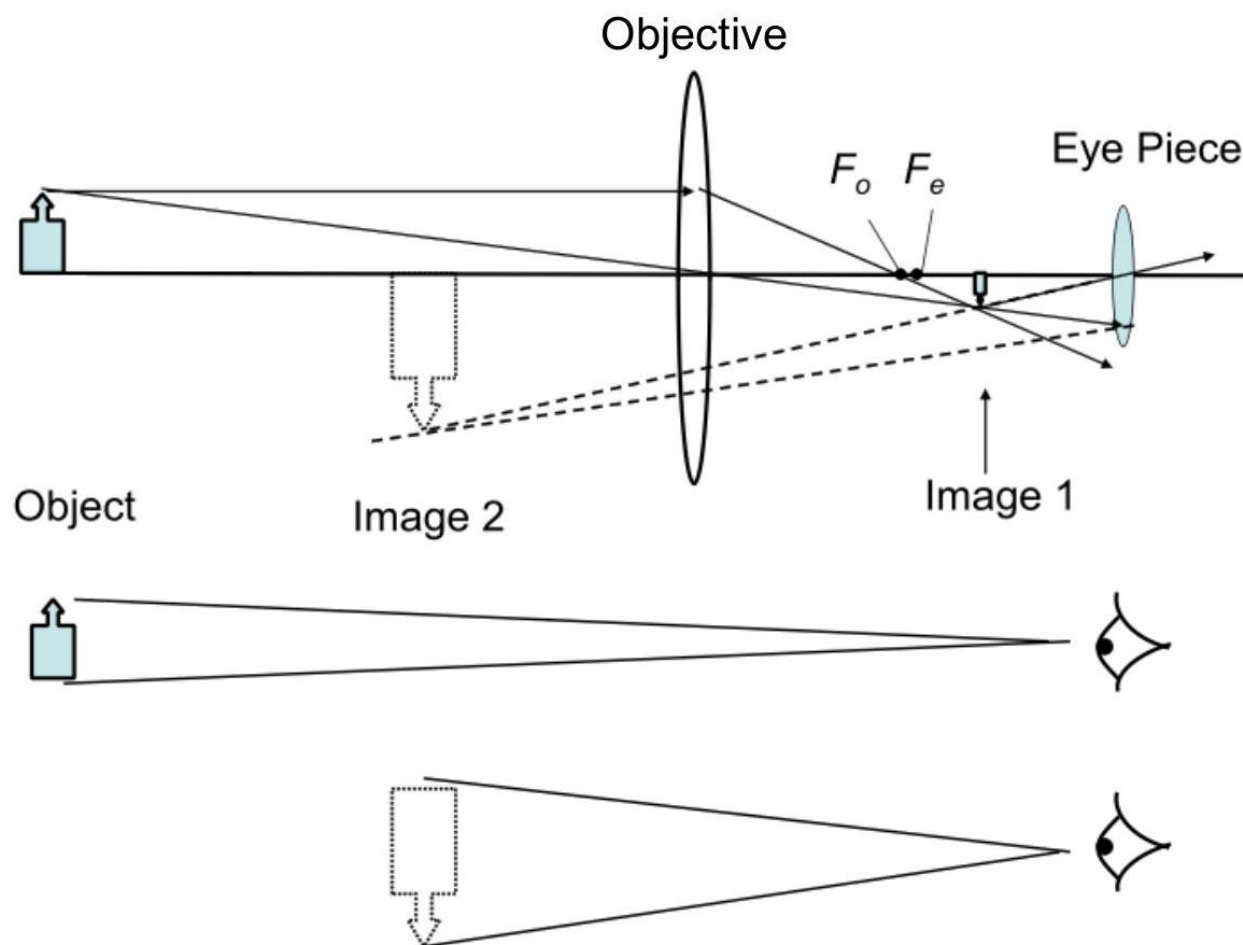
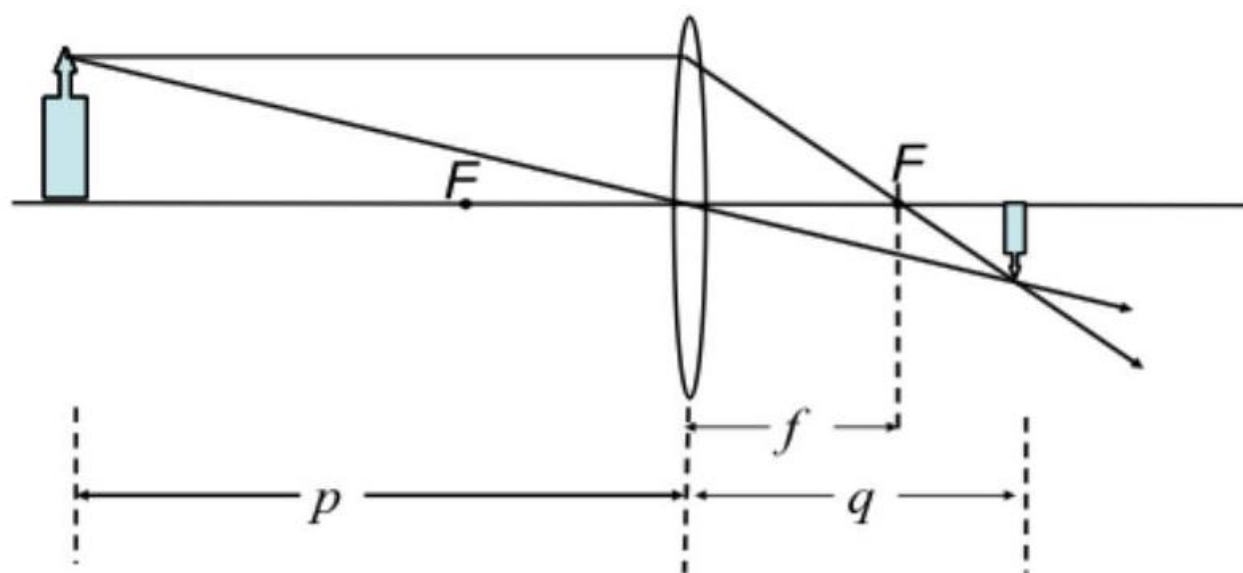


Fig. 4-Telescope and angular magnification.

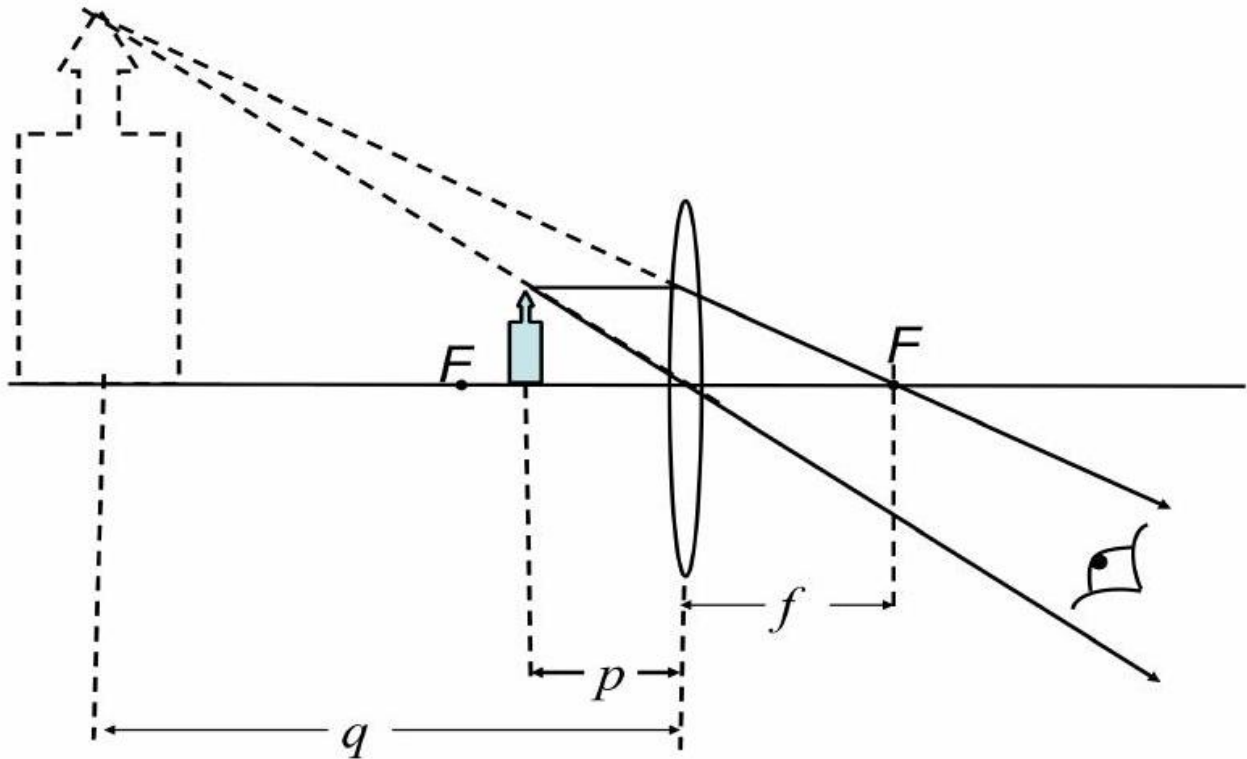


Following the above figure, place the illuminated arrow at three different distances: 30, 50, and 80 cm. Use

the white screen to view the image. For each object distance, record the object and image distances. Compute the focal lengths using $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$.

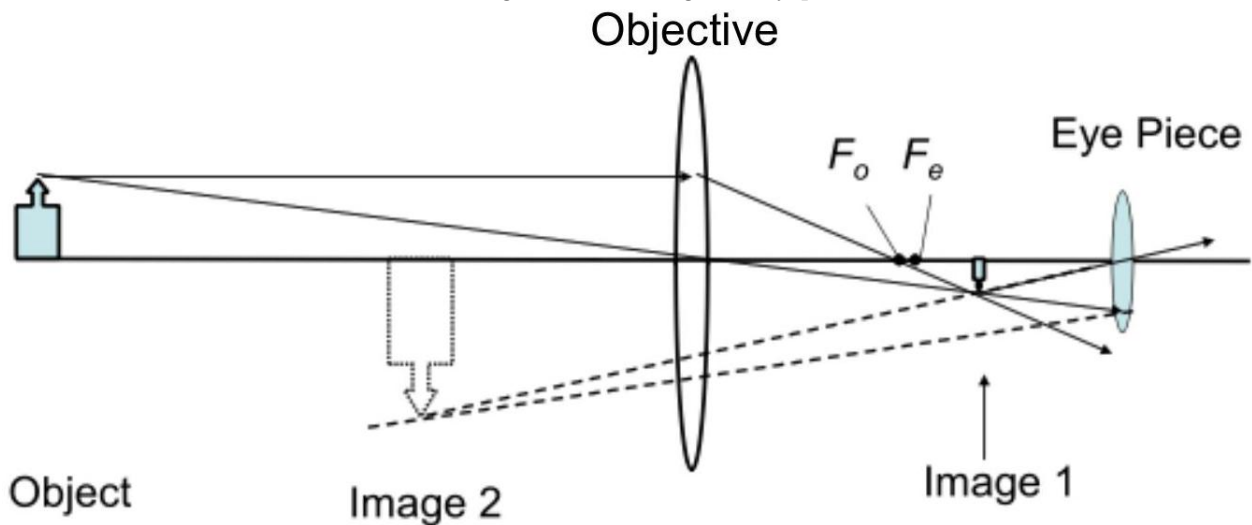
A2: Repeat step A1 using the smaller lens and compute the focal lengths.

A3: Have one lab partner hold the smaller lens at **1 cm** in front of their eye and try to view a small object through the lens, as shown in the figure below.



Starting with an object distance of 10 cm, move the object toward the lens until an enlarged image is observed. The other lab partner records the distance. Describe your observation.

B1: Construct the telescope as shown below and in in Fig. 4. Place the target screen with horizontal lines at one end of the 8-foot-long lab bench and view the lines from the other end of the bench. Place the eyepiece at one end of the optical rail and slide the carriage of the objective lens to adjust the distance between the two lenses until a clear image is seen through the eyepiece.



B2: Keep the eye at about 3 cm from the eyepiece so that the target and its image through the eyepiece

can be seen simultaneously with one eye, as illustrated in Fig. 5. Compare the distance between the lines of the target through the eyepiece and the distance viewed without the telescope to measure the angular magnification. As an example, the magnification in the Fig. 5 is 5.4.

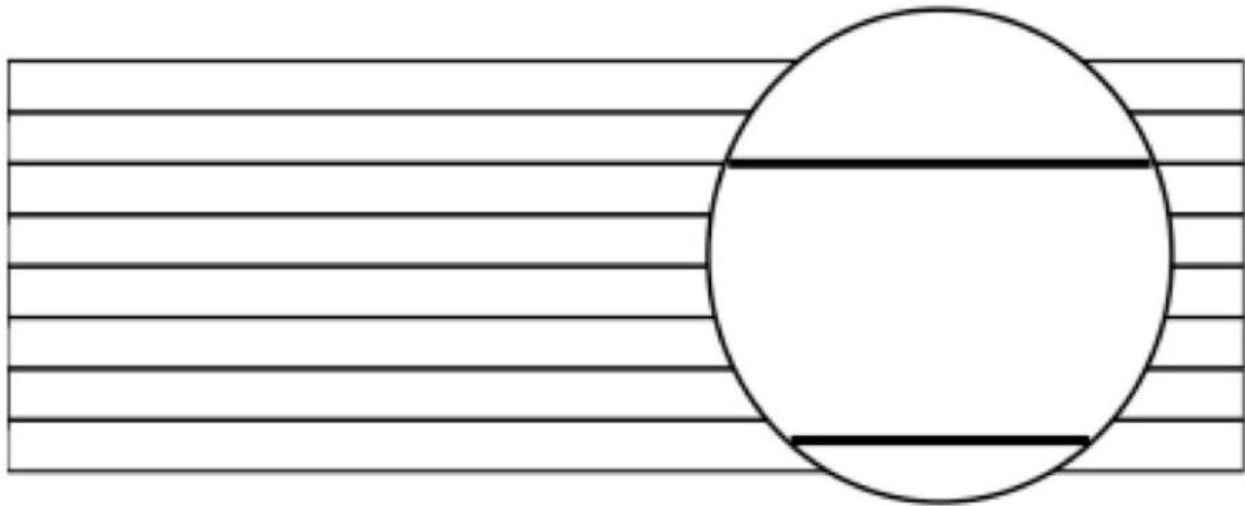


Fig. 5-Ruled target for angular magnification measurement. The magnification shown is 5.4.

Using your measurements of the two lenses focal lengths in Part A, calculate the expected angular magnification. How well do your measured and calculated values for the magnification agree with each other?

Lenses and Telescopes: ANSWER SHEET

Name: _____

Date: _____

Partner(s): _____

Instructor: _____

A1: Lens 1

Object Height: _____

Object Distance			
Image Distance			
Enlarged or reduced			
Upright or inverted			
Focal length			

Average focal length: _____

A2: Lens 2

Object Distance			
Image Distance			
Enlarged or reduced			
Upright or inverted			
Focal length			

Lenses and Telescope Classroom Lab
Average focal length: _ _ _ _

A3: Magnifier

Record the distance when a clear image is first seen as the object is moved closer.
Describe your observation. Is the image upward or inverted?

B. Angular magnification of telescope

Measured angular magnification: _ _ _ _
Calculated angular magnification: _ _ _ _
How well do your measured and calculated magnification values agree with each other?

Aim

Results

Conclusion
